- A wide range of methods for collecting data but it may be advisable to limit these to a manageable number,
- Participants should have meaningful roles in the collection and presentation of data because of the flexibility of the process and the constant reflection,
- Not every cycle will be complete.
- There may be times when it is advisable to stop mid stream and start a new cycle.

Check Your Progress 2

1) What do you understand by dialectical critical analysis?

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2) How do multiplicity of views are accommodated in action research?

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3) Match the followings:

1) Kemmis and Mc Taggart	a) Cycles of Action Research Model
2) Stringer	b) Action Research Cycle Model
3) Kurt Lewin	c) Spiral Model
4) Rial	d) Intracting Spiral Action Model

4) Enlist the features of flexibility of action research.

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20.7 STEPS INVOLVED IN CONDUCTIONING ACTION RESEARCH

Action research is a systematic process for finding the solution of the problem. It can be conducted either individually or in collaboration with others. Different models of action research have envisioned different stages. In general, following steps are involved in conducting action research:

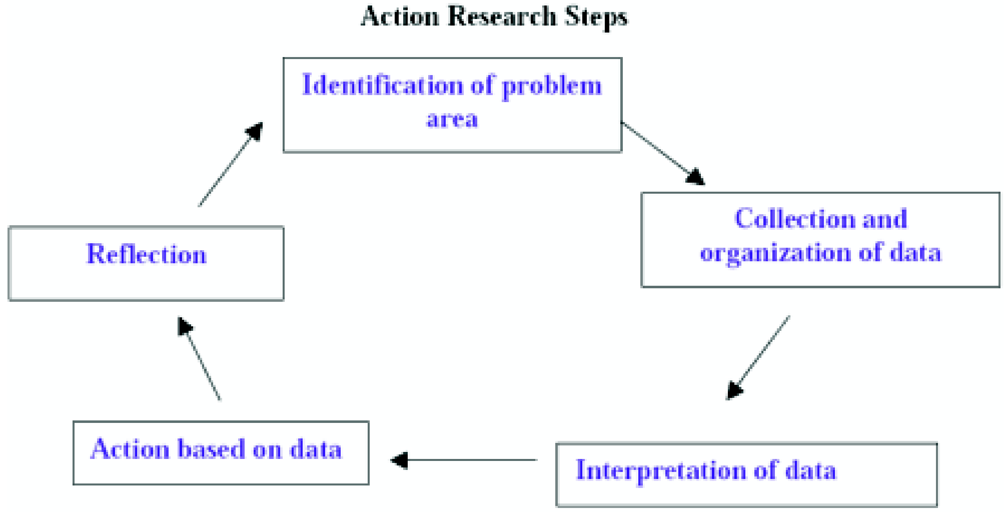

Fig. 20.11

1) **Identification of the Problem Area and Developing a Focus**

Suppose a teacher may have several questions or problems to encounter such as poor reading ability among his/her students, pronunciation problem among students, effective monitoring of the various programs and many more. Therefore, the focus of action research is on what students are experiencing or have experienced? For example, a teacher can study how to improve problem-solving skills in mathematics among the students or increase reading ability among students and so on. It is therefore crucial to choose the problem which is meaningful so that it can be solved in the stipulated time.

The need for action research arises because of perceived dissatisfaction with an existing situation. It is followed with the idea of bringing out improvement in the situation. The focus is on the following: (i) what is the cause of problem? (ii) Why is it happening? (iii) As a practitioner or a researcher, what can I do about it? (iv) Which steps can I take to solve the problem? The answers to all such questions are helpful in perceiving a problem as it exists which is a pre-requisite for undertaking any action research problem.

2) **Formulating the Problem**

Once, the problem is identified, the next step is to formulate it. The researcher tries to find causes underlying the problem along with various issues that are related to causes. These probable causes need to be stated in concise and unambiguous terms.

3) **Stating the Research Questions and Development of Propositions**

After formulating the problem, the researcher needs to state the research questions and develop a tentative theory in the form of propositions keeping in view the genesis of the problem. It is necessary to develop a conceptual and functional relationship, tentatively to understand and explain the given situation.

4) **Data Collection**

The collection of data is the most important step in deciding what action is needed for solving the problem. For example, in the school, there could be multiple sources of data, which a practitioner can use to identify causes and developing, and implementing remedial measures. These include.

- Videotapes and audio tapes,
- Report cards,
- Attendance, samples of student work, projects, performances,
- Interviews with the parents, students etc.,
- Cumulative records and Anecdotal records,
- School Diaries,
- Photos,
- Questionnaires,
- Focus groups discussions,
- Checklists,
- Observation schedules.

Select the data that are most appropriate for the study. After collecting the data, these are arranged on the basis of gender, classroom, grade level, school, etc. The practitioner may use purposive samples of students or teachers from each grade level in case of larger groups.

5) **Analysis and Interpretation of Data**

After the data has been gathered, the next step is to analyze the data in order to identify trends and themes. The qualitative data obtained can be reviewed to take out the common elements or themes and may be summarized in the suitable table formats. The quantitative data can be analyzed with the use of simple statistics such as percentages, simple frequency tables, or by calculating simple, descriptive statistics.

At this step, the data is turned into information, which can help the practitioner or the researcher in making decisions. Therefore, this stage requires maximum time. After the analysis, it becomes clear what important points do these data reveal and which important patterns or trends are emerging.

6) **Discussion and Evaluating Actions**

After the careful analysis of the data, review of current literature is done for taking decisions and necessary actions. Following points need be kept in mind while conducting the literature review:

i) Identifying topics that relate to the area of the study and would most likely yield useful information.

ii) Gather or collect research reports, research books and videotapes relating to the problem.

iii) Organise these materials for drawing inferences in the light of result of the action research study.

Suggesting a plan of action will allow the practitioner to make a change. This is well informed decision-making. It is advisable to suggest one action at a time and then observe its outcome in improving the situation.

20.8 ADVANTAGES AND DISADVANTAGES OF ACTION RESEARCH

20.8.1 Advantages

1) Action research lends itself to use in work or community situations. Practitioners, people who work as agents of change, can use it *as part of their normal activities*. Mainstream research paradigms in some field situations can be more difficult to use. Action research offers such people a chance to make more use of their practice as a research opportunity.

2) When practitioners use action research, it has the potential to increase the amount they learn consciously from their experience. The action research cycle can also be regarded as a learning cycle. Systematic reflection is an effective way for practitioners to learn.

3) Action research is usually participative. This implies a partnership between you and your clients. You may find this more ethically satisfying. For some purposes it may also be more occupationally relevant.

4) Action research is helpful in determining policy related to the problem.

20.8.2 Disadvantages

1) Excuting an action research project is more difficult than conventional research. Here a researcher takes responsibilities for change as well as for research. In addition, as with other field of research, it involves you as a researcher in more work to set it up, and you don't get any credit for that.

2) It doesn't accord with the expectations of some examiners. Deliberately and for good reasons it ignores some requirements which have become part of the ideology of some conventional research. In that sense, it is counter-cultural.

3) As a researcher, one is not exposed much about action research. Action research methodology is something that a learner has to learn almost from scratch.

4) The library work for action research is more demanding. In conventional research you know ahead of time what literature is relevant. In most forms of action research, the relevant literature is defined by the data you collect and interpret. That means that you begin collecting data first, and then go to the literature to challenge your findings.

5) Action research is more difficult to report, at least for thesis purposes. If you stay close to the research mainstream, you don't have to take the same pains to justify what you do. For action research, you are obliged to justify your overall approach.

Check Your Progress 3

1) What does motivate you to undertake action research?

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2) How do research questions enable to formulate the problem?

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3) Identify the possible sources of data for an action researcher.

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4) What are the advantages of action research over conventional research?

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5) Do you think that action research is a tough challenge before a conventional researcher? Give two reasons in support of your answer.

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20.9 LET US SUM UP

Action Research is associated with emancipatory method of critical theory paradigm approach wherein knowledge is generated in the process of knowing through doing rather through conceptualization and theorizing. In this process, it becomes difficult to make a distinction between researcher and practitioner as their activities become integrated.

Action research is a powerful tool for change and improvement at the local level. Researcher in action research aims to contribute both to the practical concerns of people in an immediate problematic situation and to further the goals of social sciences simultaneously.

Origin of Action research is traced in the work Kurt Lewin in 1940's. He was strong exponent of research action in its concern with power relations between researcher and researched and the rights of the individuals. Action research is a form of *collective* self-reflective enquiry undertaken by participants in social situations to improve the conditions of researched.

A set of six principles guide action research. These include: reflective critical analysis, dialectical critical analysis, collaborative resource, risk, plural structure, theory, practice and transformation. Different models of action research discussed in this unit are: Kemmis and McTaggart action research model, *Elliot's Action Research Model*: *O'Leary's Cycles of Research (2004),* Stringer's model (2007), Lewin's Action Research Spiral, Calhoun's Action Research Cycle, Bachman's

Action Research Spiral, Riel's Action Research Model and Hendricks's Action Research Process.

The steps followed in action research include: Identification of the Problem Area and Developing a Focus, Formulating the problem, Stating the Research Questions and Development of Propositions, Data Collection, Analysis and Interpretation of Data, Discussions and Evaluating Actions. Action research has several advantages over conventional research. However, it has several limitations as well.

20.10 KEY WORDS

Action Research : *Learning by doing***.** A group of people identifies a problem, do something to resolve it, see how successful their effects were and if not satisfied, try again.

Reflective Critical Analysis : It is the process of becoming aware of our own perceptual biases.

Dialectical Critical Analysis : It is a way of understanding the relationships between the elements that make up various phenomena in our context.

Collaborative Resource : Which is intended to mean that everyone's view is taken as a contribution to understand the situation.

Risk : It is an understanding of our own taken-for-granted processes and willingness to submit them to critique.

Plural Structure : It involves developing various accounts and critiques, rather than a single authoritative interpretation.

Collaborative : Everyone's view is taken as a contribution in understanding the situation.

Reconnaissance : Fact finding and explanation.

Planning : Identifying the issue to be changed, developing the questions and research methods to be used developing a plan related to the specific environment.

Acting : Trialling the change following your plan, collecting and compiling evidence.

Observing : Analysing the evidence and collating the findings, discussing the findings with co-researchers and/or colleagues for the interpretation.

Reflecting : Evaluating the first cycle of the process, implementing the findings or new strategy, revisiting the process.

20.11 REFERENCES

Calhoun, E. F., (1994). *How to use Action Research in Self Renewing School*. Alexandria. V A, ASLD.

Carr, W. and Kemmis, S. (1986). *Becoming Critical: Education, Knowledge and Action Research*. London: Falmer.

Ebbutt, D. (1985). *Educational Action Research: Some General Concerns and Specific Quibbles*. In R. Burgess (ed.) Issues in Educational Research: Qualitative Methods. Lewes: Falmer, 152–74.

Elliot, J. (1991). *Action Research for Educational Change*. Buckingham: Open University Press.

Ernest T. Stringer (2007). *Action Research (Third Edition)*. London: Sage Publications, 279pp (pb), ISBN: 978-1-4129-5223-1, $46.95.

G. E. Mills (2011). *Action Research: A Guide for the Teacher-Research (4th ed.)*. Pearson.

Hendricks. C. (2009). *Improving School Through Action Research; A Comprehensive Guide for Educator. (2nd ed.)*. Upper Saddle River, N J: Pearson.

Hopkins, D. (1985). *A Teacher's Guide to Classroom Research, Philadelphia*: Open University Press.

Kemmis, S. and McTaggart, R. (1992). *The Action Research Planner (third edition) Geelong*, Vic.: Deakin University Press.

Kemmis, S. and McTaggart, R. (2000). "*Participatory Action Research*", in N.K. Denzin and Y.S. Lincoln (eds) Handbook of Qualitative Research (2nd ed.). Sage, CA, pp. 567–605.

McNiff, J., and Whitehead, J. (2002). *Action Research: Principles and Practice (second edition)*. London: RoutledgeFalmer.

O'Leary, Z. (2004). *The Essential Guide to Doing Research*. London: Sage Publications.

Richard. Winter and Carol Munn-Giddings(2001). *A Handbook for Action Research in Health and Social Care*. by Routledge, New York NY 10017.

Selener, D. (1997). *Participatory Action Research and Social Change*. New York: Cornell Participatory Action Research Network.

Zuber-Skerritt, O. (ed.) (1996). *New Directions in Action Research, London*; Falmer Press.

20.12 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) See Section 20.1
2) See Section 20.1
3) See Section 20.2
4) See Section 20.3

Check Your Progress 2

1) See Sub-section 20.4.2
2) See Sub-section 20.4.5
3) See Section 20.6
4) See Sub-section 20.6.4

Check Your Progress 3

1) See Section 20.7
2) See Section 20.7
3) See Section 20.7
4) See Sub-section 20.8.1
5) See Sub-section 20.8.2